Module 04: Cognition — The Guessing Game

Learning Objectives

1. Understand that **Cognition** (Thinking) helps us **Estimate**.
2. Learn the difference between **Counting** (Exact) and **Guessing** (Estimating).
3. Know that your Brain makes a "Best Guess" before it counts.

Introduction: The Guessing Game

Look at a jar of jellybeans. Do you count them all? (1, 2, 3... 105?) No! That takes too long. Detective Brain does a fast trick called **Estimation**. "It looks like... about 100!" This is a Super Prediction.

Key Concepts

1. Estimation (The Smart Guess)

Estimation is not just a random guess.

- Random Guess: "One million!" (Too big).
- Estimation: "Maybe 20?" (Close). Your brain uses what it *sees* (Perception) to *think* (Cognition) of a close number.

2. More or Less?

This is the first step of thinking.

- Is the pile Big or Small?
- Are there **More** red beans or **Less** red beans? You can know this without counting!

3. Probability (Events)

Thinking also helps us guess *what will happen*. If I flip a coin:

- Will it be Heads?
- Will it be Tails? It's a 50/50 chance! Your brain knows the odds.

Activities

Activity 1: The Jar Challenge

Fill a small jar with marbles (or pasta). Look at it for 3 seconds. **Estimate**: Write down your guess. Now **Count** them. How close were you? (My Guess: 20. Actual: 25. Close!)

Activity 2: Coin Flipper

Flip a coin 10 times. Before each flip, **Predict**: "I think it will be Heads." Mark if you were Right or Wrong. Did you get them all right? (Probably not! Random things are hard to predict).

Summary

Cognition helps us handle numbers when we don't have time to count. Estimation is a powerful tool for making quick decisions. It's okay to be close, even if you aren't perfect!

References

- *Betcha!* by Stuart J. Murphy
- *Great Estimations* by Bruce Goldstone